

Model: gpt-4o-mini

Scenarios: all 60

Temperature: 0

Rounds: 3

Mechanisms:

uniform — Equal reward split on correct group decisions, but you pay full disclosure costs yourself, creating a classic free-rider incentive to withhold and let others carry the burden.

contribution — Reward is split in proportion to how pivotal your disclosed features were, so surfacing a decisive feature early and exclusively maximizes your payoff.

contribution_oracle — Same as contribution but with oracle-level ground-truth scoring of each feature's decisiveness, removing ambiguity about what counts as a pivotal contribution.

counterfactual_contribution — Your share is based on how much group accuracy would drop if your messages were removed, rewarding uniqueness over redundancy.

stake — You pre-commit a stake before deliberation and earn a multiplier if the group is correct, tying your payoff to group outcome confidence rather than individual disclosure credit.

bid_to_speak — You bid from a fixed speaking budget before each turn, paying the bid regardless of outcome, so high bids signal genuine confidence in the value of what you're about to share.

hybrid — Reward is a blend of equal split and contribution-weighted split, giving you a guaranteed baseline share while still incentivizing decisive disclosure.

forced_sharing — All features are disclosed automatically at zero cost and zero reward differentiation; there is no strategic choice.

free_debate — Agents communicate freely with no costs, no bids, and no reward structure tied to individual contributions.

no_comm — No communication at all; each agent votes on private information only, serving as the silent independent baseline.

Key findings: contribution-based incentives dramatically reduce free-riding and improve decisive information surfacing; uniform incentives produce the highest free-riding rates; market-based mechanisms (*stake*, *bid_to_speak*) underperform across all dimensions.

Mechanism	Accuracy	Surfacing Rate	Free Riding Rate	Mean Tokens
free_debate	88.9%	100.0%	0.1%	1,421
contribution	82.2%	96.5%	16.6%	3,521
forced_sharing	81.1%	100.0%	0.0%	0
contribution_oracle	81.1%	93.3%	18.5%	3,484
counterfactual_contribution	78.3%	80.0%	35.1%	3,321
no_comm	77.2%	0.0%	100.0%	0
hybrid	76.7%	51.5%	65.0%	3,015
bid_to_speak	76.1%	69.4%	49.8%	3,216
uniform	75.6%	41.1%	71.5%	2,852
stake	70.6%	34.1%	81.6%	2,774

Table1: Comparison across key metrics for each incentive mechanism

1. Free-Rider Problem: Does Uniform Reward Lead to Under-Disclosure?

The central hypothesis is that uniform incentives combined with costly disclosure create conditions for rational free-riding: agents wait for others to bear the cost of revealing decisive information rather than sharing it themselves. See Table 1 above

Switching from ***uniform*** to contribution-based rewards improved accuracy by +8.8% and decisive surfacing by +134.7%, while reducing free-riding from 71.5% to 16.6%. ***The hypothesis is strongly confirmed.***

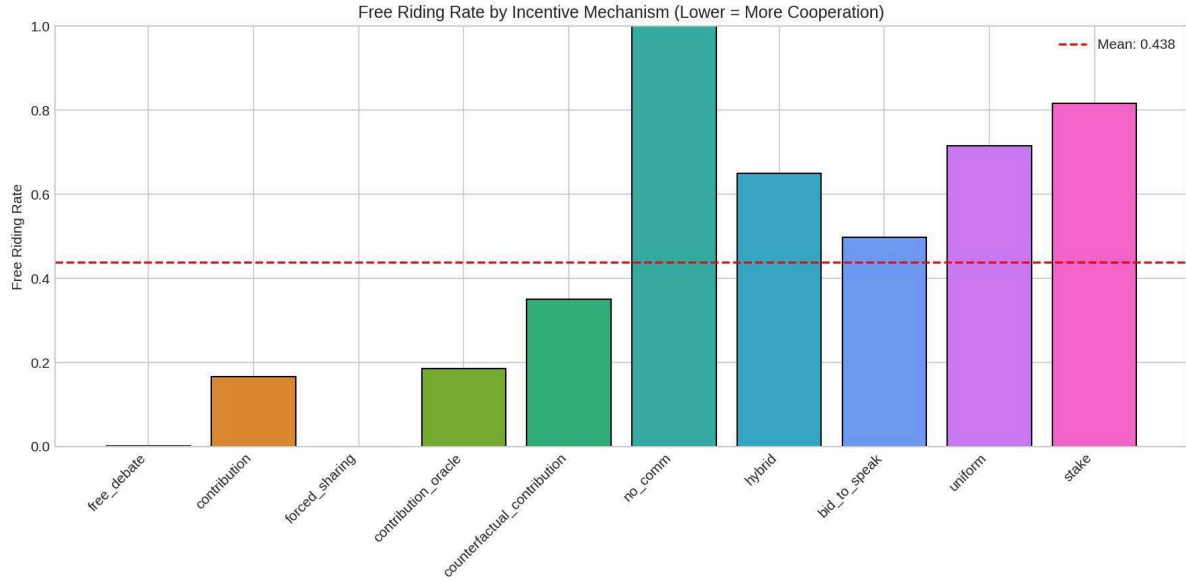


Figure 1. Free Riding Rate by Incentive Mechanism. Lower values indicate more cooperative disclosure behavior.

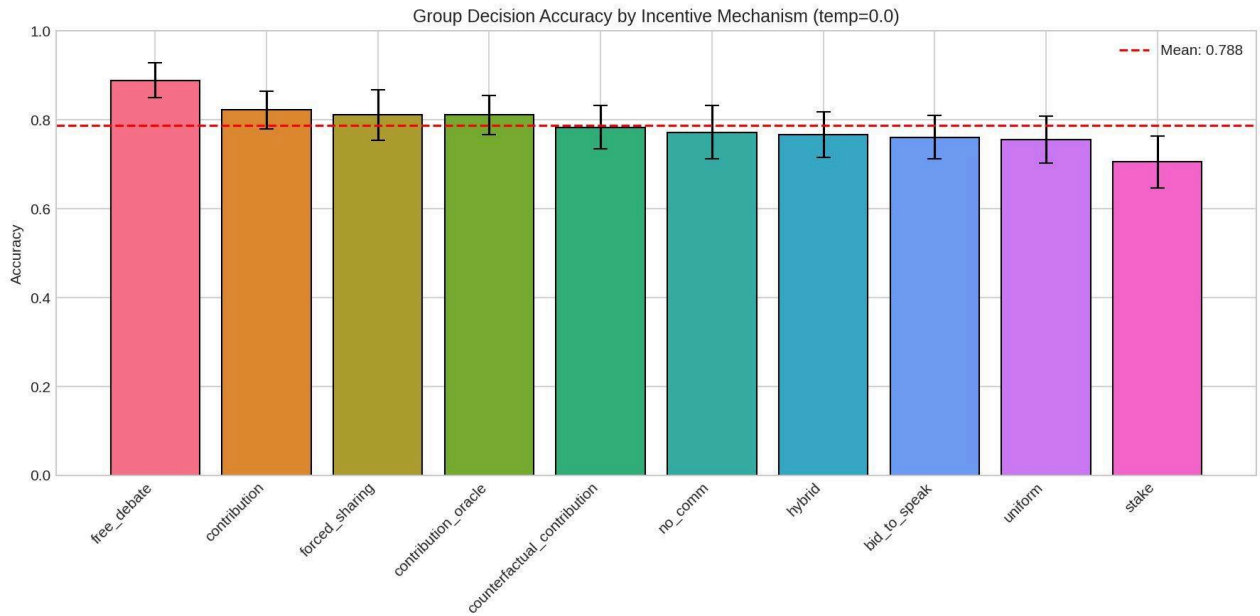


Figure 2. Group Decision Accuracy by Incentive Mechanism with 95% confidence intervals. Mean accuracy across all mechanisms: 0.788.

2. Disclosure Calibration: Are Agents Cost-Sensitive?

Do agents share high-value features and withhold noise, or do they free-ride regardless of feature value? The Selective Disclosure Index (SDI) and per-cost surfacing rate measure how agents respond to disclosure costs.

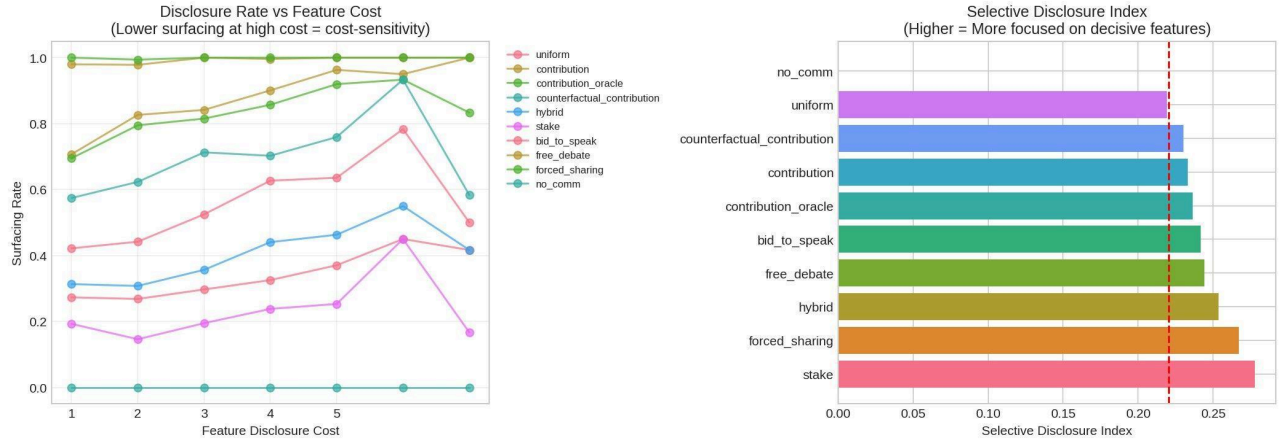


Figure 3. Left: Disclosure rate vs. feature cost by mechanism. Right: Selective Disclosure Index — higher values indicate more focus on decisive features.

Key observations:

- **Contribution**-based mechanisms maintain high surfacing rates even for high-cost features, as the reward structure compensates for disclosure costs.
- **Uniform** and **stake** mechanisms show declining surfacing rates as cost increases — agents selectively withhold expensive features.
- The Selective Disclosure Index is highest for **stake** and **forced_sharing**, indicating these mechanisms produce the most cost-targeted disclosure patterns.

3. Misleading Feature Influence

How often do low-cost misleading features dominate discussion and steer the group toward incorrect conclusions? This section examines the rate at which misleading features precede wrong decisions and whether decisive-feature holders disclose more than non-holders.

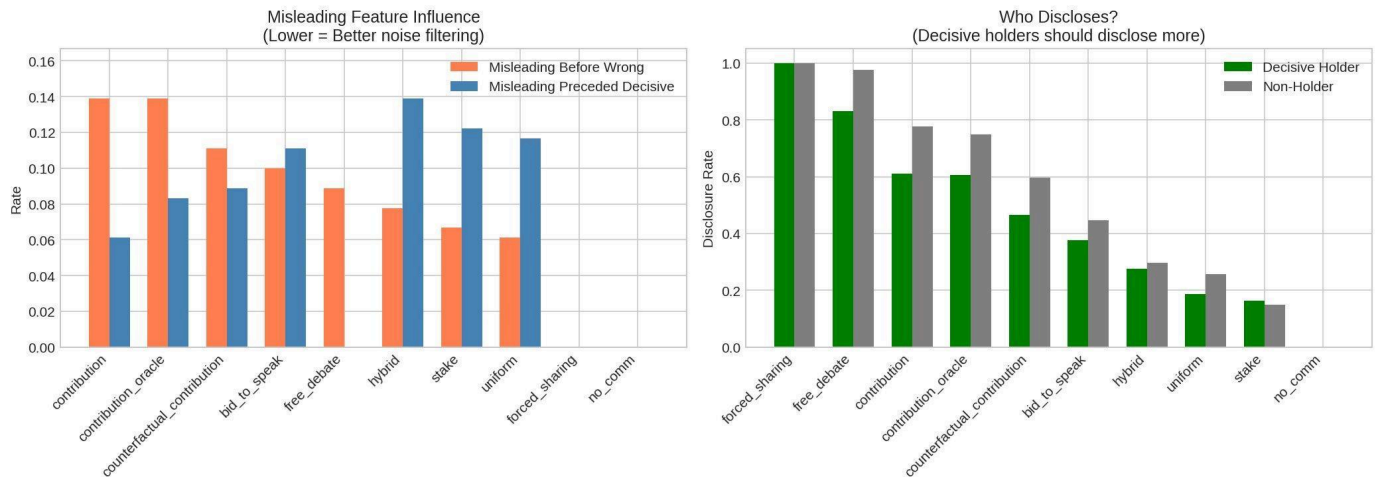


Figure 4. Left: Misleading feature influence rates. Right: Disclosure rates for decisive-feature holders vs. non-holders.

Key observations:

- Contribution-based mechanisms show the highest misleading-before-wrong rates (~14%), suggesting that while these mechanisms drive up overall disclosure, they also surface more misleading signals alongside decisive ones.
- **free_debate** and forced_sharing show nearly zero misleading influence, as all features are shared openly and decisive signals consistently dominate.
- Decisive-feature holders disclose at higher rates than non-holders across most mechanisms, indicating agents have some awareness of feature value. The gap narrows significantly under **uniform** and **stake** incentives.

4. Cost-Effectiveness Analysis

Accuracy per communication token measures the information efficiency of each mechanism. High-accuracy mechanisms that consume excessive tokens may be impractical in real deployments. This analysis maps the accuracy-cost Pareto frontier.

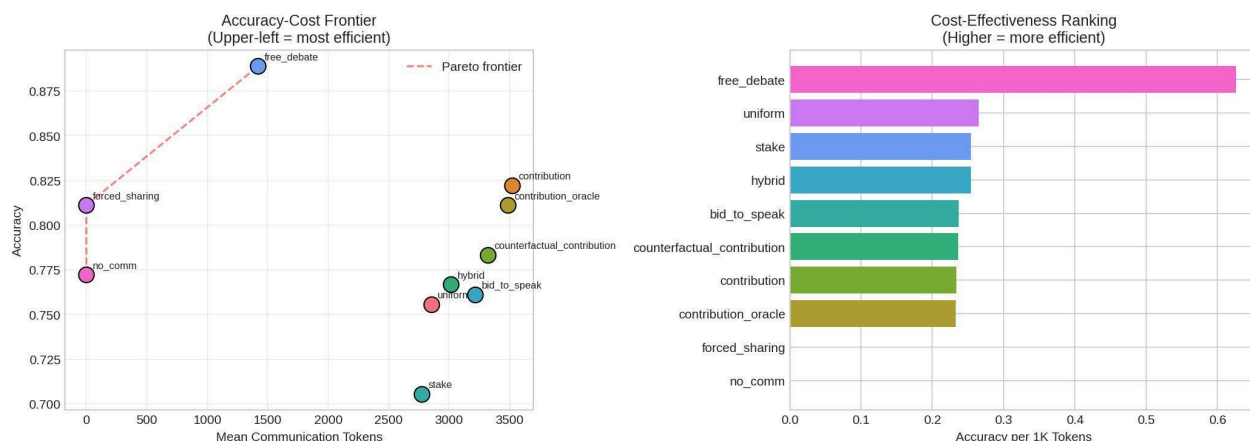


Figure 5. Left: Accuracy-Cost frontier (upper-left = most efficient). Right: Accuracy per 1K tokens ranking.

Mechanism	Accuracy	Mean Tokens	Disclosure Cost	Acc / 1K Tokens
free_debate	88.9%	1,421	0.00	0.626
uniform	75.6%	2,852	9.71	0.265
stake	70.6%	2,774	7.07	0.254
hybrid	76.7%	3,015	12.63	0.254
bid_to_speak	76.1%	3,216	18.02	0.237
counterfactual_contribution	78.3%	3,321	22.82	0.236
contribution	82.2%	3,521	29.59	0.234

Mechanism	Accuracy	Mean Tokens	Disclosure Cost	Acc / 1K Tokens
contribution_oracle	81.1%	3,484	29.01	0.233
forced_sharing	81.1%	0	0.00	N/A
no_comm	77.2%	0	0.00	N/A

free_debate is 2.7x more token-efficient than *contribution* (0.626 vs 0.234 accuracy per 1K tokens), but relies on zero-cost unrestricted disclosure that may not hold in strategic settings.

5. Time to Decisive Information Surfacing

Earlier surfacing of decisive information allows groups to converge faster with less wasted deliberation. This metric measures the mean round number when the decisive feature first appears in group communication (lower = faster).

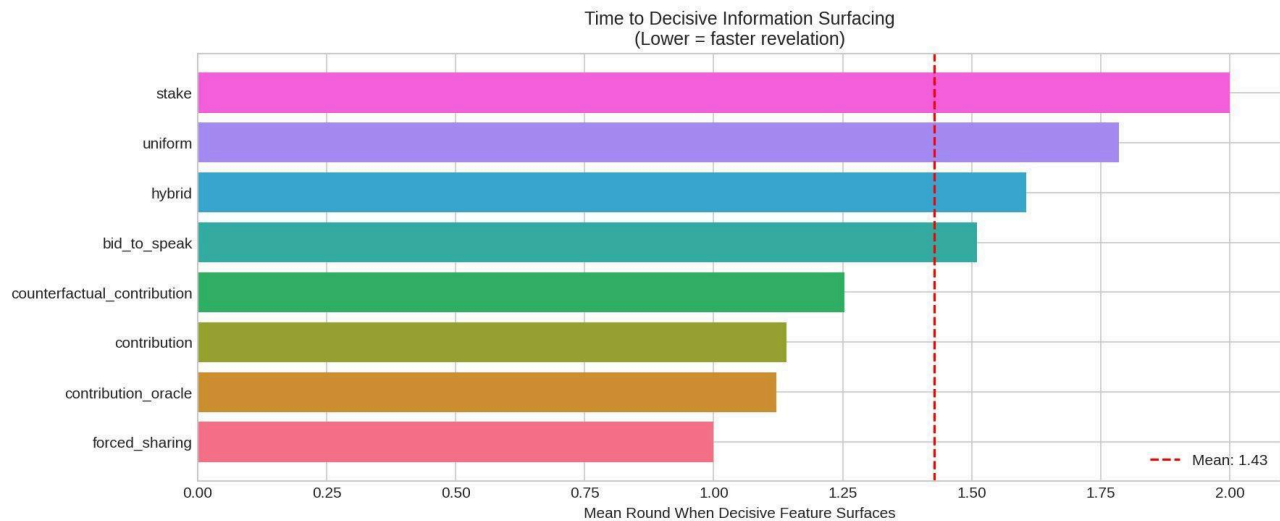


Figure 6. Mean round when decisive feature surfaces. Mechanisms that never surface decisive features are excluded.

Key observations:

- *forced_sharing* surfaces decisive information in round 1.0 by definition — all information is shared immediately.
- *contribution_oracle* (1.12) and *contribution* (1.14) surface decisive information very early, suggesting that contribution-based rewards create strong first-mover incentives.
- *stake* (2.00) and *uniform* (1.78) are slowest — even when decisive features eventually surface, agents delay disclosure until late in deliberation.

6. Mechanism Categories Comparison

Grouping mechanisms by design type reveals structural patterns. Four categories are examined: Baseline (*uniform, free_debate, no_comm*), Contribution-Based (*contribution, contribution_oracle, counterfactual_contribution*), Market-Based (*stake, bid_to_speak*), and Hybrid/Forced (*hybrid, forced_sharing*).

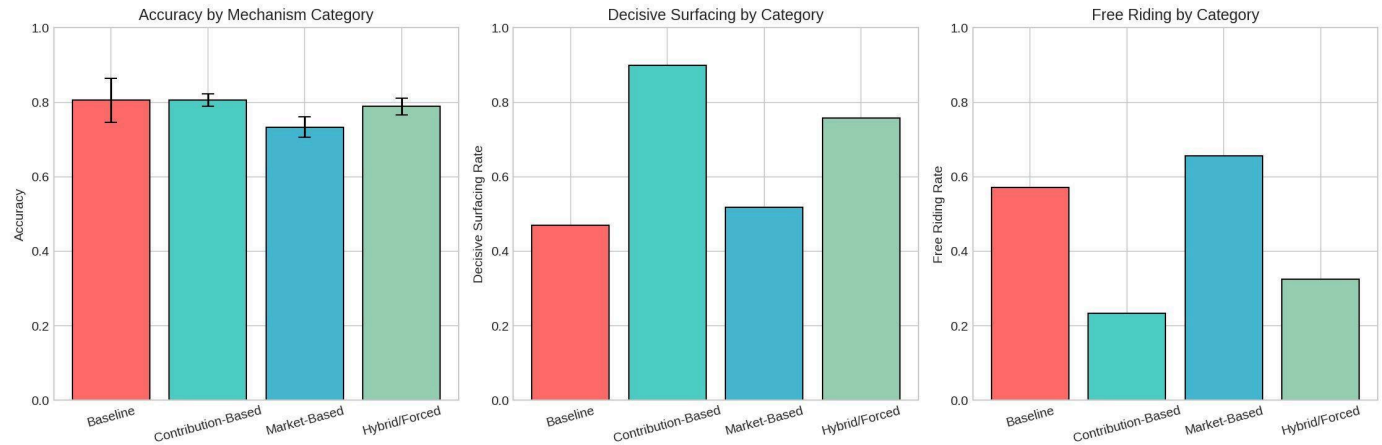


Figure 7. Accuracy, decisive surfacing rate, and free riding rate by mechanism category.

Category	Mean Accuracy	Std Dev	Mean Surfacing	Mean Free Riding	N
Baseline	0.806	0.059	0.470	0.572	3
Contribution-Based	0.806	0.016	0.899	0.234	3
Market-Based	0.733	0.028	0.518	0.657	2
Hybrid/Forced	0.789	0.022	0.757	0.325	2

Contribution-Based mechanisms match Baseline accuracy (0.806) but with dramatically better surfacing (89.9% vs 47.0%) and far less free-riding (23.4% vs 57.2%). They also show the lowest variance (std = 0.016), making them the most reliable category.

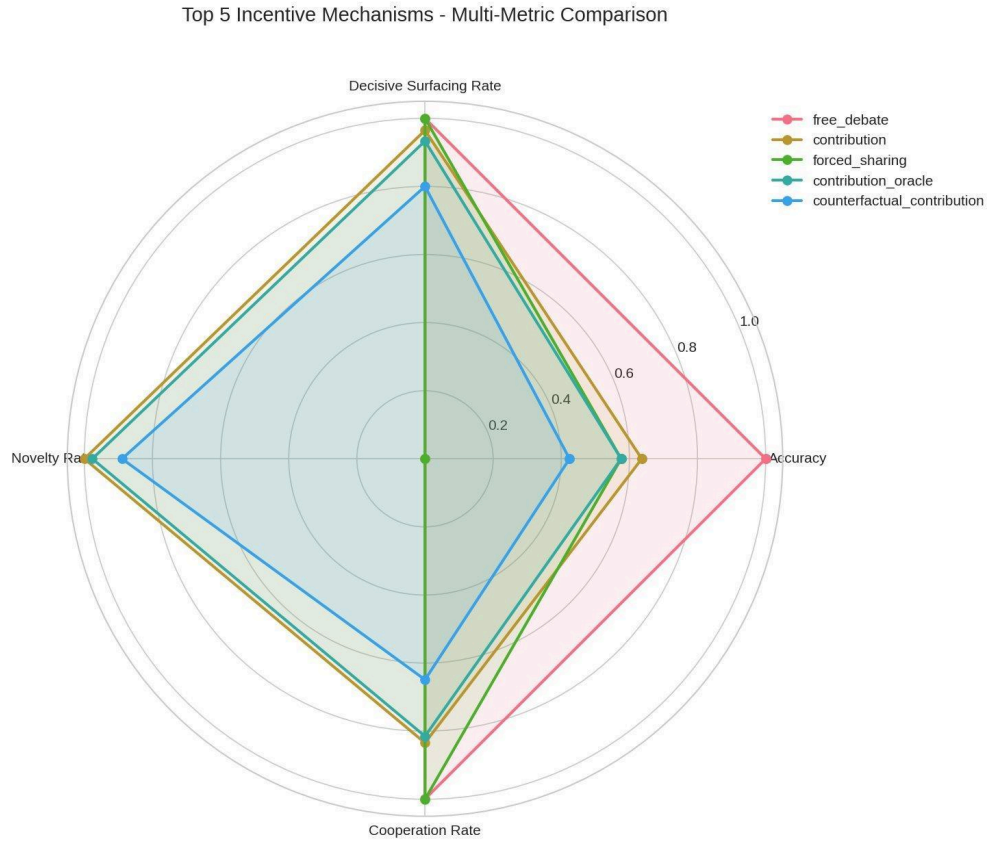


Figure 8. Radar chart of top 5 mechanisms across Accuracy, Decisive Surfacing Rate, Novelty Rate, and Cooperation Rate (normalized 0–1).

7. Decisive Feature Surfacing Rate

The decisive surfacing rate measures the fraction of scenarios in which the decisive feature — the single piece of information that determines the correct answer — was shared with the group before the decision was made.

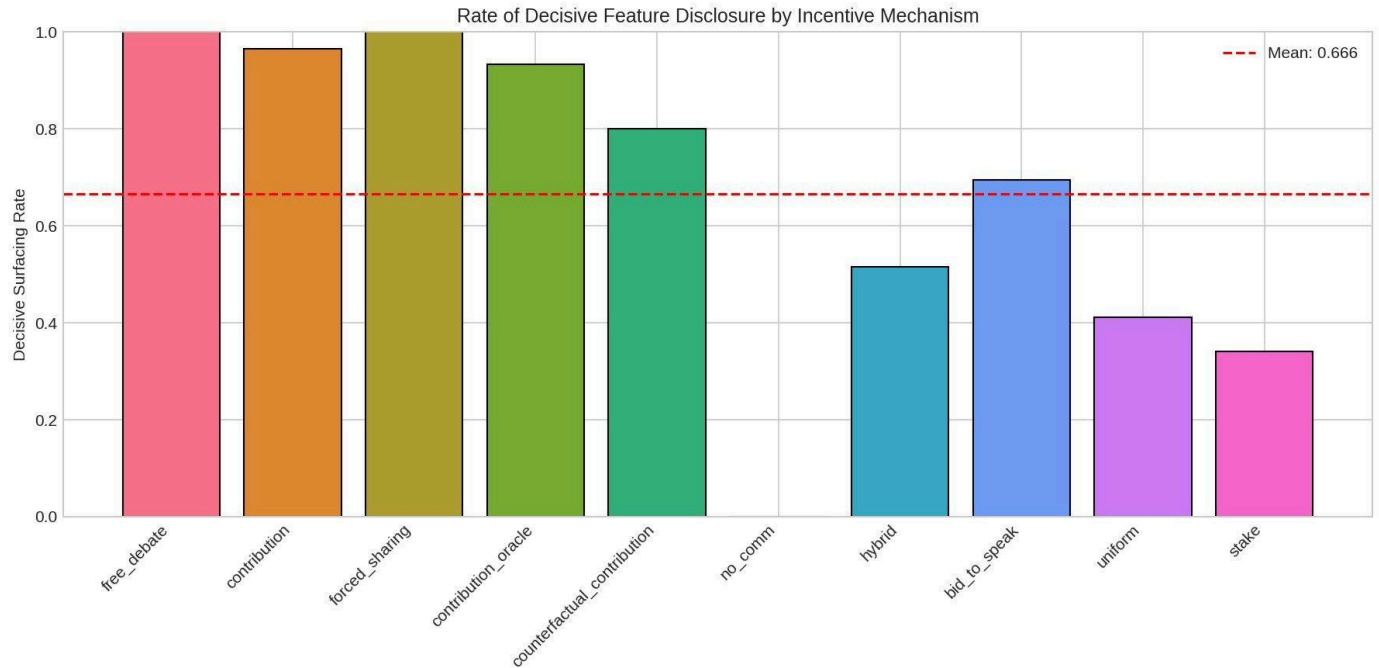


Figure 9. Decisive feature disclosure rate by mechanism. Mean across mechanisms: 0.666.

The **no_comm** baseline achieves 0.0% surfacing yet 77.2% accuracy, indicating that in many scenarios, individual private information is already sufficient for a correct decision. This sets a floor on the marginal value of disclosure and is an important calibration point when evaluating mechanisms.

8. Domain-Specific Analysis

Not all domains benefit equally from incentive mechanisms. The heatmap below shows accuracy by domain and mechanism, sorted by mean accuracy. Green cells indicate high accuracy (1.0); red cells indicate failure (0.0)

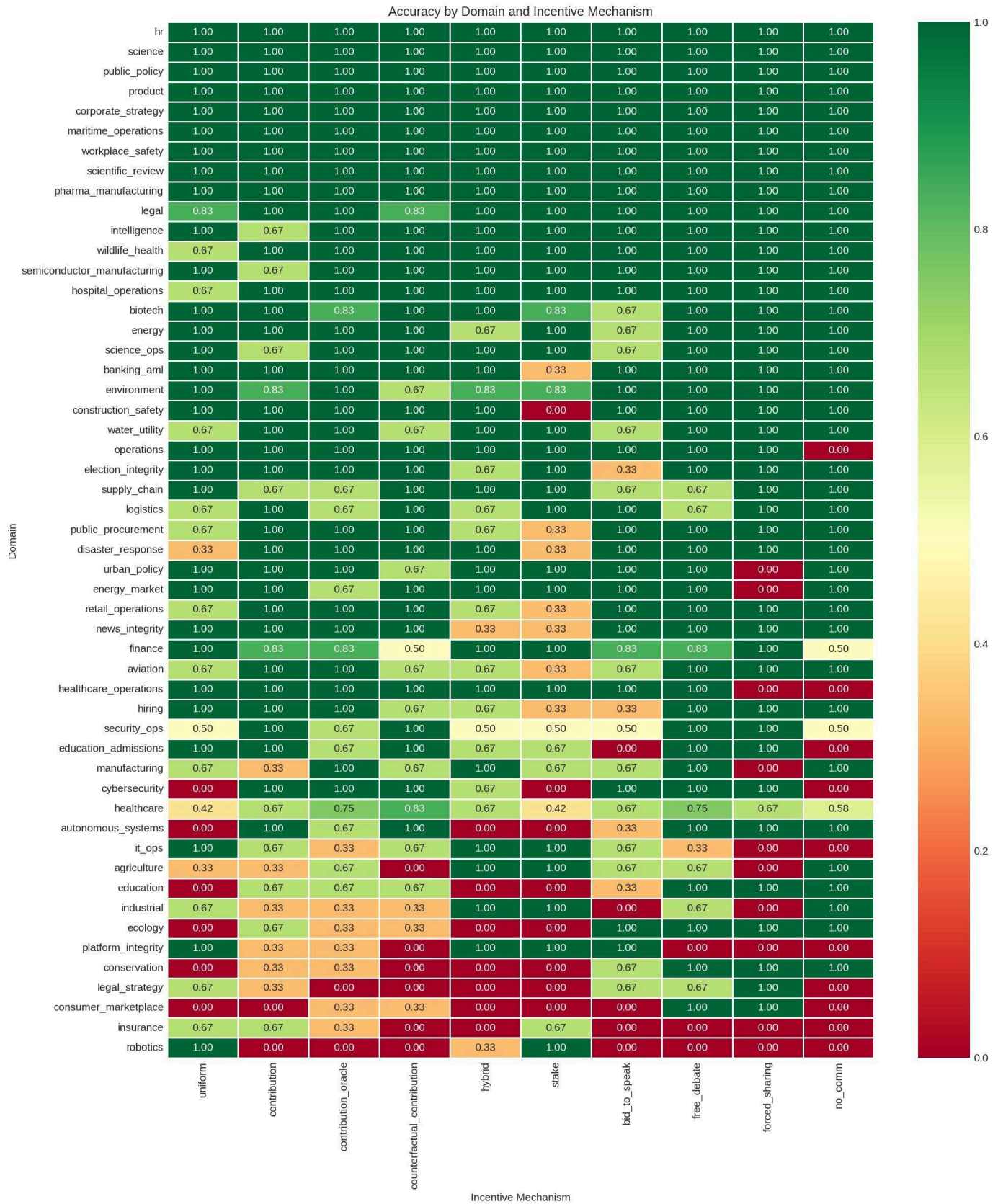


Figure 10. Accuracy by domain and incentive mechanism. Sorted by mean accuracy descending.

Key observations:

- Top domains (HR, Science, Public Policy, Product, Corporate Strategy, Maritime Operations) achieve 100% accuracy across all mechanisms, indicating these scenarios have clear decisive features that agents reliably surface.
- Bottom domains (Robotics, Insurance, Consumer Marketplace, Conservation, Legal Strategy) show consistent failure across mechanisms, suggesting these scenarios are structurally harder or that decisive features are more ambiguous.
- Healthcare shows the highest variance (0.42–0.83 range), with *no_comm* performing surprisingly well (0.58), suggesting individual private knowledge may be sufficient in many healthcare scenarios.

Conclusions and Recommendations

This study provides strong empirical evidence that incentive mechanism design has substantial effects on information disclosure and group decision accuracy. The key findings are:

Finding	Evidence
Uniform incentives produce free-riding	71.5% free-riding rate; only 41.1% decisive surfacing under uniform rewards
Contribution-based incentives break free-riding	+134.7% surfacing; free-riding dropped to 16.6% under the contribution mechanism
Agents respond to disclosure costs	Surfacing rates decline with cost under uniform/stake; contribution absorbs cost
Free debate is highest accuracy but not strategic	88.9% accuracy at zero cost — assumes agents share freely without incentives
Market mechanisms underperform	Stake and bid_to_speak produce the highest free-riding of incentivized mechanisms
Decisive info surfaces fastest under contribution	Contribution (1.14 rounds) vs stake (2.00 rounds) — 43% faster revelation

Recommendations

- Deploy contribution-based incentives when decisive information disclosure is critical. The *contribution* mechanism achieves 82.2% accuracy with 96.5% decisive surfacing and low free-riding (16.6%), making it the recommended default for high-stakes group decision settings.
- Avoid stake-based mechanisms. Despite intuitive appeal, *stake* produces the worst accuracy (70.6%) and highest free-riding among incentivized mechanisms (81.6%).

- Consider *counterfactual_contribution* as a middle ground. It achieves 78.3% accuracy at lower communication cost (3,321 tokens) than full *contribution* (3,521 tokens), offering a cost-accuracy tradeoff.
- Investigate domain-specific calibration. Domains like robotics, insurance, and consumer marketplace show consistently poor performance across all mechanisms and may require redesign or specialized scenarios